

TECHNICAL & PRODUCT ANALYSIS REPORT

Pocket Lawyer

A Source-Grounded Digital Legal Access Platform for Sri Lanka

Prepared from the implemented repository

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Live deployment: pocketlawyer.site

1. Executive Summary

Pocket Lawyer is a full-stack legal technology platform built for the Sri Lankan legal context. It unifies a curated law library, a source-grounded AI legal assistant, a verified lawyer directory, subscription and payment operations, content publishing, and a comprehensive administration layer into a single, coherent product.

The platform addresses a practical access-to-justice problem: legal material in Sri Lanka is often difficult to locate, difficult to interpret, and disconnected from the professionals who can act on it. Pocket Lawyer narrows this gap by retrieving supporting legal material, citing its sources, refusing to answer beyond the evidence it can support, and routing users to verified lawyers when a matter requires professional judgment.

157 API Route Handlers	27 Persisted Domain Models	105 React Modules	122 Discovered Test Functions
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Note: the figures above are static repository counts and are reported as implementation evidence, not as a certification that every code path is production-ready.

2. Problem Statement and Product Ideology

The product is guided by a single governing principle: technology should make the law easier to navigate while preserving a clear boundary between legal information and professional legal advice. This principle is expressed through six design commitments.

- Access before complexity — a fragmented research journey is reduced to a clear path: discover, understand, verify and escalate.
- Evidence before fluency — legal claims are tied to retrieved acts, sections and trusted sources rather than generated from memory alone.
- Human counsel remains essential — the AI assistant supports orientation; lawyers handle high-risk, uncertain or fact-specific matters.
- Local context matters — knowledge base, lawyer discovery, and payment methods are designed specifically around Sri Lanka.
- Privacy is a product feature — private-mode AI conversations are not persisted; standard sessions remain user-manageable.
- Governance is built in — prompt versions, usage, feedback, failures, document review, permissions and audit trails are all visible to administrators.

The platform is positioned as an educational and legal-information service rather than a substitute for legal representation. Its credibility is intended to come from citation transparency, explicit limitations, controlled refusal behavior, and a direct route to qualified professionals.

3. User Roles

Role	Primary Capabilities
Public visitor	Explore articles, FAQs, the law library, advocate listings and subscription plans.
Registered user	Use the AI assistant, manage chat sessions, access the law library, and maintain a profile and subscription.
Lawyer	Apply for verification, publish content, and manage a discoverable professional profile.
Administrator	Govern users, lawyers, AI configuration, documents, content, payments, support and platform permissions.

4. Feature Overview

4.1 AI Legal Assistant

- Conversational Q&A with session history, rename and delete controls.
- Grounded answers with document- and section-level citations.
- Private mode that avoids conversation persistence.
- Answer feedback capture and plan-based monthly AI usage limits.
- Voice input via NVIDIA Riva / Whisper transcription.
- Response handling in English, Sinhala and Tamil.

4.2 Sri Lankan Law Library

- Browse, search, filter and view featured law books.
- Protected document viewing and download paths.
- Streamed AI document summaries via Server-Sent Events.
- Follow-up questions against cached summaries.
- Authentication- and subscription-aware access gating.

4.3 Advocate Discovery

- Verified lawyer directory with detailed profiles.
- Location and professional-information filtering.
- AI-to-lawyer recommendation for uncertain or high-risk cases.
- Lawyer profile management and professional publishing tools.

4.4 Identity and Accounts

- Email OTP signup and login flows.
- Google Sign-In with server-side token verification.
- JWT access/refresh sessions with revocation support.
- TOTP two-factor authentication for administrator accounts.
- Profile, avatar, password, theme and account-deletion controls.

4.5 Subscriptions and Payments

- Weekly, monthly, quarterly and annual subscription tiers.
- Managed plans, public offers and percentage-based discounts.
- PayHere recurring checkout with signed notifications and cancellation APIs.
- Stripe checkout/webhook path and administrator-managed bank transfers.
- Payment history, status controls, invoices and cancellation flows.

4.6 Legal Publishing

- Public legal blog with article detail pages.
- Lawyer-authored posts with cover-image upload.
- Legal news management with rich-text editing and duplication.
- FAQ, contact and support communication channels.

4.7 Lawyer Onboarding

- OTP-verified application and submission process.
- Professional information and identity-card upload.
- Private storage for sensitive verification documents.
- Administrator review, approval/rejection and lifecycle controls.

4.8 Experience Layer

- Responsive interfaces for public, user, lawyer and admin roles.
- Dark/light themes with Framer Motion transitions.
- English, Sinhala and Tamil locale dictionaries.
- In-app notifications and audience-targeted announcements.
- Lazy-loaded routes, loading states and reusable dialog components.

5. Administration and Governance

Operational control is treated as a core part of the product rather than an auxiliary set of CRUD screens. The administration surface governs AI behavior, editorial content, legal documents, billing, staff permissions and platform health.

Domain	Scope
Users and lawyers	Search/edit users, bans, bulk actions, session revocation, login history, application review, verified-profile management.
AI governance	Runtime configuration, prompt CRUD/testing/version restore, usage reporting, answer feedback, failure analysis.
Knowledge operations	Law-book CRUD, file replacement, processing jobs, page/chunk review, chunk rebuilding, reindexing, vector status.
Commercial operations	Plans, offers, payments, bank transfers, verification/rejection, exports, invoices.
Content and communications	Blogs, news, media, settings, support replies/resolution, targeted announcements.
Security and accountability	Admin bootstrap/invites, roles, permissions, account security, audit logs, system health.

6. AI and Retrieval-Augmented Generation Architecture

The platform's answer pipeline connects a React AI workspace to a FastAPI orchestration layer, a ChromaDB vector store, and NVIDIA-hosted AI models for embedding, reranking, chat generation and speech transcription.

6.1 Ingestion and Indexing Pipeline

Stage	Description
1. Ingest	PDFs, scans, images, text, Markdown and DOCX sources enter the processor.
2. Extract	Native text extraction, or Tesseract/AWS Textract OCR for scanned material.
3. Structure	Section-aware chunks preserve title, section, category, page and source metadata.
4. Embed	NVIDIA passage embeddings are generated in concurrent batches.
5. Index	ChromaDB stores cosine-similarity vectors; SHA-256 manifests skip unchanged files.
6. Retrieve	The user question is embedded and matched against top semantic candidates.
7. Rerank	An NVIDIA cross-encoder reranker improves evidence precision.
8. Generate	The Nemotron model answers only from labeled evidence and attaches citations.
9. Fail safely	Weak, unsafe or unsupported requests trigger refusal and lawyer escalation.

6.2 Grounding and Multilingual Behavior

Prompt rules require source markers for legal claims and preserve official names and citations. The assistant operates in English, Sinhala and Tamil. When local retrieval is insufficient, an allow-listed fallback can query official and trusted Sri Lankan legal web sources for additional, rerankable evidence.

7. Application Architecture

The system separates interface, business logic, persistence and AI concerns into distinct, testable layers.

Layer	Technologies	Role
Presentation	React SPA	Public site, AI workspace, laws, advocates and role-specific dashboards.
Client state	Context + hooks	Authentication, chat, theme, settings and dialog state.
API	FastAPI + Pydantic	Typed REST, uploads, SSE, dependency access and OpenAPI.

Layer	Technologies	Role
Services	Python domain layer	Auth, users, lawyers, content, payments, notifications, documents.
Persistence	SQLAlchemy + PostgreSQL	Relational business, governance and audit records.
Knowledge	ChromaDB	Embedded legal chunks and retrieval metadata.
Storage	Amazon S3 / local	Avatars, legal PDFs, images and private identity documents.
External services	NVIDIA, PayHere, Stripe, Google, SMTP, Tavily	AI, speech, payments, identity, email and trusted-source search.
Delivery	Vite, Docker, Uvicorn, Nginx, AWS EC2	Build pipeline, container runtime and reverse-proxied deployment.

The data model spans users, authentication events, lawyer applications and profiles, chat sessions and messages, AI usage/feedback/failure logs, laws and summaries, document processing jobs, blogs, news, media, FAQs, contact and support records, notifications, announcements, plans, offers, subscriptions, payments, prompts, runtime settings, admin roles and permissions, invites, and audit logs.

8. Technology Stack

8.1 Product Application

Layer	Technologies	Role
Languages	JavaScript (ES modules), HTML5, CSS3	Browser application and component styling.
UI	React 18, React DOM	Component architecture and client rendering.
Routing	React Router DOM 6	Public, authenticated and role-specific SPA routes.
Motion	Framer Motion	Transitions, reveals and interaction animation.
Styling	Tailwind CSS 3, PostCSS, Autoprefixer	Responsive utilities, theme variants, CSS processing.
Build	Vite 5, React plugin	Development server, proxy, code splitting and builds.
Frontend quality	ESLint, React/Hooks plugins	Static checks and hook rules.
Backend language	Python 3.12	API, domain logic, AI orchestration and indexing.
Web API	FastAPI 0.115, Uvicorn	Async server, dependency injection, REST, files, SSE, docs.
Validation	Pydantic 2, pydantic-settings	Typed schemas and environment configuration.
ORM	SQLAlchemy 2 (async)	Persistence, transactions and query composition.
Database	PostgreSQL (Amazon RDS), asyncpg, psycopg2-binary	Managed relational store with async runtime and sync migrations.
Migrations	Alembic	Versioned schema evolution.
Networking	HTTPX, Tenacity, orjson	External calls, retry patterns, efficient JSON.
Reverse proxy	Nginx	TLS termination, static asset serving and request routing to Uvicorn.
Hosting	Amazon EC2	Application server instance running the containerized backend.
Managed database	Amazon RDS for PostgreSQL	Production PostgreSQL instance with automated backups and maintenance.

8.2 AI, Data and Integrations

Layer	Technologies	Role
AI platform	NVIDIA API Catalog / NIM-compatible endpoints	Hosted chat, embedding and reranking.
Chat model	Llama 3.3 Nemotron Super 49B v1.5	Grounded multilingual answer generation.
Embeddings	llama-nemotron-embed-1b-v2	Passage and query vectors.
Reranking	llama-nemotron-rerank-1b-v2	Higher-precision evidence selection.
Speech	NVIDIA Riva client / Whisper endpoint	Voice transcription.
AI client	OpenAI Python SDK	Async interface to NVIDIA-compatible APIs.
Vector database	ChromaDB, HNSW cosine	Persistent legal semantic index.
Tokenization	tiktoken	Usage estimation and accounting.
Documents	pypdf, pdfplumber	PDF text and page extraction.
OCR	Tesseract, pytesseract, pdf2image, Pillow, Poppler	Scanned PDF/image processing.
Cloud OCR	AWS Textract (boto3)	Optional OCR alternative.
Storage	Amazon S3 (boto3), local fallback	Public/private file uploads.
Payments	PayHere, Stripe, bank transfer	Recurring checkout, webhooks and manual settlement.
Identity	Google Identity, google-auth, Authlib	Google Sign-In and validation.
Email	SMTP, aiosmtplib, Jinja2	OTP and transactional messaging.
Web evidence	Tavily + allow-listed HTTP fetch	Optional trusted legal source fallback.

9. Security, Quality and Delivery

9.1 Authentication and Sessions

- JWT access and refresh tokens; Passlib/bcrypt password hashing.
- Email OTP and Google ID-token verification.
- Encrypted TOTP secrets using Fernet.
- Session revocation and login history.

9.2 Authorization and Privacy

- Roles for free, Pro, lawyer and administrator users.
- Fine-grained admin roles and permissions.
- Private chat mode that avoids persistence.
- Sensitive identity documents stored via private storage paths.

9.3 AI Safety and Observability

- Source-required prompting with weak-evidence refusal.
- Unsafe or unsupported request handling with lawyer escalation.
- Token limits, usage records, feedback and failure logs.
- Prompt versioning and runtime AI configuration.

9.4 Payment Integrity

- PayHere signature, amount and currency validation.
- Stripe webhook signature verification and duplicate protection.
- Safe replacement and cancellation of recurring plans.
- Administrator verification for bank transfers.

9.5 Testing and Delivery

Pytest and pytest-asyncio cover backend flows, with focused RAG tests covering chunking, manifests and review artifacts. Backend tests run against SQLite/aiosqlite rather than a live PostgreSQL instance. Frontend quality gates include ESLint and Vite build checks. Delivery assets include Alembic migrations, a multi-stage Dockerfile with a non-root user and health check, backup tooling, and an EC2/Nginx/systemd deployment plan.

9.6 Live Deployment

The platform is deployed on Amazon Web Services. The application runs on an EC2 instance behind Nginx, which handles TLS termination and reverse-proxies requests to the Uvicorn/FastAPI backend and the built React frontend. The relational database runs on Amazon RDS for PostgreSQL, separating compute and data

tiers and providing managed backups and maintenance. The platform is publicly reachable at pocketlawyer.site.

Repository inventory: 157 FastAPI route decorators, 27 persisted model modules, 105 JSX modules, 114 backend test functions and 8 RAG-specific test functions. These are static repository counts and do not represent a claim that every path is production-certified or that all tests were executed for this report.

10. Assessment and Positioning

Pocket Lawyer demonstrates full-stack engineering across a responsive frontend, a typed asynchronous API, relational persistence and multiple third-party integrations. Its applied-AI component covers ingestion, OCR, legal-aware chunking, embedding, vector search, reranking and citation-based answer presentation.

Security is addressed through OTP verification, Google identity integration, JWT session management, TOTP two-factor authentication, role-based access control and private-data handling for sensitive documents. Platform-level thinking is evident in the billing, notification, support, content-operations, audit and health-monitoring capabilities that extend the product beyond a single AI feature.

The project is best assessed as a responsible legal-access platform: the AI assistant helps users locate and understand relevant material, the interface discloses the sources behind its answers, and the product provides a direct path to verified human legal expertise. The core engineering achievement is the integration of these responsibilities — access, evidence, privacy and governance — into a single, operable system.

Note: Pocket Lawyer provides general legal information and platform services. It is not a substitute for advice from a qualified legal professional.